

Course Code:B20BS2101																							
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)								R 20															
II B. Tech I Semester Regular Examinations																							
NUMERICAL METHODS & VECTOR CALCULUS																							
(Common to CE, CSE, EEE & IT)																							
Time: 3 Hrs.						Max. Marks:70																	
Answer any one Question from Each Unit																							
All questions carry equal Marks																							
	UNIT-I							CO	KL	M													
1.a)	Determine a real root of the equation $x \log_{10} x = 1.2$ by Regula-falsi method and correct to two decimal places							CO1	K2	7													
b)	Determine the cube root of 41 using Newton-Raphson method							CO1	K2	7													
(OR)																							
2. a)	Determine Newton's forward difference interpolation formula find Y (3), from the following table							CO1	K2	7													
<table><tr><td>X</td><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>Y</td><td>7</td><td>11</td><td>14</td><td>18</td><td>24</td><td>32</td></tr></table>							X				0	5	10	15	20	25	Y	7	11	14	18	24	32
X	0	5	10	15	20	25																	
Y	7	11	14	18	24	32																	
b)	Using Lagrange's interpolation formula find Y (10) from the following table							CO1	K2	7													
<table><tr><td>X</td><td>5</td><td>6</td><td>9</td><td>11</td></tr><tr><td>Y</td><td>12</td><td>13</td><td>14</td><td>16</td></tr></table>							X				5	6	9	11	Y	12	13	14	16				
X	5	6	9	11																			
Y	12	13	14	16																			
UNIT-II																							
3.a)	Evaluate $\int_0^1 x^3 dx$ with five subintervals by trapezoidal rule							CO2	K3	7													
b)	Evaluate $\int_0^2 \frac{dx}{x^3+x+1}$ by using Simpsons 1/3 rd rule with $h = 0.25$							CO2	K3	7													
(OR)																							
4. a)	Employ Taylor's method to obtain approximate value of y at $x=0.2$ for the differential equation $\frac{dy}{dx} = 2y + 3e^x, y(0) = 0$							CO2	K3	7													
b)	Evaluate $y(0.2)$ using Runge-Kutta 4 th order method, given $\frac{dy}{dx} = \frac{y^2-x^2}{y^2+x^2}, y(0) = 1.$							CO2	K3	7													
UNIT-III																							
5.a)	Apply change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dx dy$							CO3	K3	7													
b)	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar coordinates							CO3	K3	7													
(OR)																							

6. a)	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$	CO4	K3	7
b)	Determine the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$	CO4	K3	7
UNIT-IV				
7. a)	Obtain the directional derivative of $\phi = xy + yz + zx$ at A in the direction of AB where $A = (1,2,-1)$, $B = (5,6,8)$.	CO5	K2	7
b)	Determine the values of a and b such that the surface $ax^2 - byz = (a+2)x$ and $4x^2y + z^3 = 4$ cut orthogonally at $(1,-1,2)$.	CO5	K2	7
(OR)				
8. a)	Show that the vector $(x^2 - yz)\bar{i} + (y^2 - zx)\bar{j} + (z^2 - xy)\bar{k}$ is irrotational and find its scalar potential.	CO5	K2	7
b)	Determine $Curl \bar{F}$ and $div \bar{F}$ for $\bar{F} = x^2y\bar{i} - 2xz\bar{j} + 2yz\bar{k}$	CO5	K3	7
UNIT-V				
9.a)	Determine the work done in moving a particle once round the circle $x^2 + y^2 = 9$ in the xy- plane by the force $\bar{F} = (2x - y - z)\bar{i} + (x + y - z^2)\bar{j} + (3x - 2y + 4z)\bar{k}$	CO6	K3	7
b)	Evaluate the line integral by Stokes's theorem for the vector function $\bar{F} = y^2\bar{i} + x^2\bar{j} + (z+x)\bar{k}$ and C is the triangle with vertices $(0,0,0)$, $(1,0,0)$ and $(1,1,0)$.	CO6	K3	7
(OR)				
10.	Verify Green's theorem in the plane For $\oint_C [(3x^2 - 8y^2)dx + (4y - 6xy)dy]$, where C is boundary of the region defined by $y = \sqrt{x}$, $y = x^2$	CO6	K3	14

Course Code: B20IT2101					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech I Semester Regular Examinations					
DATA STRUCTURES					
(Common to AIDS & IT)					
Time: 3 Hrs.				Max. Marks:70	
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a)	Define data structure. Discuss different types of data structure their implementations applications.	CO1	K2	7
	b)	Explain the recursive merge sort algorithm to sort the following elements: 12, 25, 5, 9, 1, 84, 63, 7, 15, 4, 3.	CO1	K2	7
OR					
2	a)	Write Recursive Binary Search algorithm. Search element 12 in the given list using Recursive Binary Search algorithm 2, 5, 12, 15, 23, 28, 36, 39, 45	CO1	K2	7
	b)	Rearrange following numbers using quick sort: 10, 6, 3, 7, 17, 26, 56, 32, 72	CO1	K3	7
UNIT-II					
3	a)	Explain the procedure to evaluate postfix expression. Evaluate the following postfix expression 7 3 4 + - 2 4 5 / + * 6 / 7 +	CO2	K2	7
	b)	Discuss the algorithms for push and pop operations on a stack	CO2	K2	7
OR					
4	a)	Give the structure of Queue ADT. Explain the operations in it.	CO2	K2	7
	b)	Convert the given infix expression $A+B^{\wedge}C+(D^{\wedge}E/F)^{\wedge}G$ into its postfix expression, and evaluate the same using stack. Here A=3, B=5, C=2, D=7, E=4, F=1, G=8.	CO2	K3	7
UNIT-III					
5	a)	Write an algorithm for representing the polynomial $6x^0 + 4x^3 - 2x + 10$ using linked lists.	CO2	K3	7
	b)	Write an algorithm to insert new node at the beginning and at the end of a singly linked list.	CO2	K3	7
OR					
6	a)	Write an algorithm to push and pop an element from linked stack	CO2	K3	7
	b)	Write an algorithm to delete an element anywhere from doubly linked list.	CO2	K3	7
UNIT-IV					
7	a)	Explain Heap sort algorithm. Create Heap for the following elements and then sort them. (13, 102, 405, 136, 15, 105, 390, 432, 28, 444)	CO3	K3	7
	b)	Develop a binary search tree resulting after inserting the following integer keys 49, 27, 12, 11, 33, 77, 26, 56, 23, 6. (i) Check whether the tree is almost complete or not? (ii) Determine the height of the tree (iii) Write post order and pre-order traversals	CO3	K3	7
OR					
8	a)	How to represent binary tree using arrays and linked list?	CO3	K2	7
	b)	Construct an AVL tree by inserting the following elements successively C O M P U T E R	CO3	K2	7
UNIT-V					

9	a)	Explain Warshall's algorithm to find transitive closure of a graph with a suitable example.	CO3	K2	7
	b)	What is minimum cost spanning Tree? Explain the process of finding the minimum spanning tree with suitable example.	CO3	K2	7
OR					
10	a)	Explain Depth First Search algorithms in detail.	CO3	K2	7
	b)	Explain Dijkstra's algorithm with suitable example.	CO3	K2	7

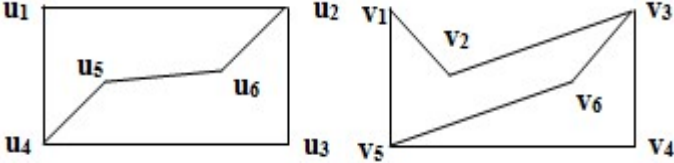
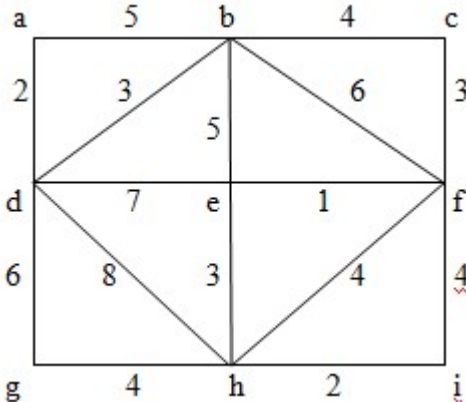
Course Code: B20IT2102					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech I Semester Regular Examinations					
DATABASE MANAGEMENT SYSTEM					
(For IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT - I					
1.	a).	Explain the problems associated with conventional file processing system.	CO1	K2	8M
	b).	Explain different levels of abstraction offered by DBMS with an example	CO1	K2	6M
OR					
2.	a).	Draw a neat diagram of the structure of DBMS and explain the functions of various components of DBMS.	CO1	K2	8M
	b).	Explain about Centralized and Client-Server architectures of DBMS.	CO1	K2	6M
UNIT - II					
3.	a).	A University has several departments. Each department has several instructors and one among them is the HOD. Each instructor teaches multiple courses. Each department offers several courses. A student can enrol for several courses offered by different departments. Considering above description, develop a complete E-R diagram for the University database.	CO2	K3	7M
	b).	Consider the following tables: Students(Std_ID:String, S_Name:String, Date_of_Birth: Date) and Enrolled(Std_ID String, Course_ID String, Grade Char(1)) Answer the following queries in SQL. i) Write DDL statement to create Students table. ii) Display student ID, name of those students who got 'A' grade in the course "C01". iii) Find IDs and names of students who born in the year 2000.	CO3	K4	7M
OR					
4.	a).	Explain clearly, the concept of weak entity set with a semantic example of your choice.	CO2	K3	7M
	b).	Consider the following tables: Suppliers(Supp_ID: Integer, S_Name: String, Address:String) Parts(P_ID: Integer, Pname: String, Colour: String) Catalog(Supp_ID: Integer, P_ID: Integer, Cost: Real). Answer the following queries in SQL. i) Write DDL statements to create Suppliers and Catalog tables with key constraints and referential constraint. ii) Find IDs and names of Suppliers who supply some "Red" colour part. iii) Find the details of suppliers whose names are 5 characters long and begin with character 'A'.	CO3	K4	7M
UNIT - III					
5.	a).	Draw an E-R diagram of a binary relationship with 1..M mapping cardinality and translate the complete diagram into a collection of	CO2	K3	6M

		relations. Specify all integrity constraints. Write SQL statements to create the resulting tables.			
	b).	Consider the following schema. Students (Std_ID: String, S_Name: String, Dept: String, GPA: Real) Courses (C_ID: String, C_Name: String, Credits: Integer, Offered_by_Dept: String) Enrolled (Std_ID: String, C_ID:String, Grade: Character). Answer the following queries in SQL. i) For each course offered by CSE department, find the total number of enrolments. ii) Find the sum of credits of all courses enrolled by student "S01". iii) Find the courses that have at least 10 enrolments. iv) Applying outer join, find the count of enrolments for each course offered.	CO3	K4	8M
OR					
	a).	Consider E-R diagram of a weak entity set and translate the complete diagram into a collection of relations. Specify all integrity constraints. Write SQL statements to create the resulting tables.	CO2	K3	6M
6.	b).	Consider the following schema. Students (Std_ID: String, S_Name: String, Dept: String, GPA: Real) Courses (C_ID: String, C_Name: String, Credits: Integer, Offered_by_Dept: String) Enrolled (Std_ID: String, C_ID:String, Grade: Character). Answer the following queries in SQL. i) Write a correlated query to find IDs and names of students who are enrolled for course "C01". ii) Create a view named "GoodGrades" which contain Std_ID, S_Name and C_ID of enrolments for which the grade is "A". iii) Display Students table in the descending order of names of students. iv) Find the IDs and names of students who enrolled for both the courses "C02" and "C03".	CO3	K4	8M
UNIT – IV					
7.	a).	Consider the schema R(A,B,C,D,E,G) and the list of functional dependencies $F = \{A \rightarrow BC, EC \rightarrow D, D \rightarrow A, G \rightarrow E\}$. Determine all candidate keys of R.	CO4	K4	7M
	b).	Explain with an example, join dependency and fifth normal form.	CO4	K3	7M
OR					
8.	a).	Find the best normal form satisfied by the relation R(A,B,C,D,E) with set of FDs $F = \{BC \rightarrow D, AC \rightarrow BE, B \rightarrow E\}$ Decompose R into the next higher normal form.	CO4	K4	7M
	b).	Explain with an example, multivalued dependency and fourth normal form	CO4	K3	7M
UNIT - V					
9.	a).	Explain briefly ACID properties of a transaction	CO5	K2	7M
	b).	Explain the three phases of ARIES recovery algorithm	CO5	K2	7M
OR					
10.	a).	Explain search and insert operations on a B+ tree index structure.	CO5	K2	7M
	b).	Explain with an example, hash based indexing	CO5	K2	7M

Course Code: B20IT2103					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech I Semester Regular Examinations					
PYTHON PROGRAMMING					
(Common to AIDS & IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT-I					
1	a)	Explain about input validation loops and nested loops with examples	CO1	K3	7
	b)	Write a Python program to calculate the amount payable if money has been lent on simple interest. Principal or money lent = P, Rate of interest = R% per annum and Time = T years. Then Simple Interest (SI) = (P x R x T)/ 100. Amount payable = Principal + SI. P, R and T are given as input to the program.	CO1	K4	7
OR					
2	a)	Explain about explicit conversion with examples.	CO1	K2	7
	b)	Explain about precedence of all operators in Python.	CO1	K2	7
UNIT-II					
3	a)	Define Python string padding functions? Explain with examples	CO2	K2	7
	b)	Illustrate if, if-else, if-elif-else Statements with examples.	CO2	K3	7
OR					
4	a)	Explain about data encryption in Python.	CO2	K2	7
	b)	Presume that a ladder is put upright against a wall. Let variables length and angle store the length of the ladder and the angle that it forms with the ground as it leans against the wall. Write a Python program to compute the height reached by the ladder on the wall for the following values of length and angle: i) 16 feet and 75 degrees ii) 20 feet and 0 degrees iii) 24 feet and 45 degrees iv) 24 feet and 80 degrees	CO2	K3	7
UNIT-III					
5	a)	Write a Python program to create three dictionaries, then create one dictionary that will contain the other three dictionaries.	CO3	K3	7
	b)	Describe Python list/Array methods? Explain.	CO3	K2	7
OR					
6	a)	Discuss about importing module from a package.	CO3	K2	7
	b)	Explain about anonymous or Lambda function with merits and demerits	CO3	K2	7
UNIT-IV					
7	a)	Explain about structuring classes with inheritance and polymorphism.	CO4	K2	7
	b)	Illustrate manipulating file pointer using seek with suitable example.	CO4	K3	7
OR					
8	a)	Demonstrate the case study of an ATM using classes.	CO4	K3	7
	b)	Explain about reading numbers from a file using Python program.	CO4	K2	7

UNIT-V					
9	a)	Defining clean-up actions .	CO5	K2	7
	b)	Illustrate Entry fields for the input and output of text with example.	CO5	K3	7
OR					
10	a)	Describe user Defined exception with example.	CO5	K2	7
	b)	Define Scrolling list boxes With example .	CO5	K2	7

Q.No.	Questions	M	CO	KL
UNIT-I				
1.a)	Establish that $\{((p \vee q) \rightarrow r) \wedge (\neg p)\} \rightarrow (q \rightarrow r)$ is a tautology	7	CO1	K2
b)	Establish that the following argument is valid by using the rules of inference If Clifton does not live in France, then he does not speak French. Clifton does not drive a Datsun If Clifton lives in France, then he rides a bicycle Either Clifton speaks French, or he drives a Datsun Hence, Clifton rides a bicycle	7	CO1	K3
(OR)				
2.a)	Verify that the following argument is valid by using the rules of inference, quantifiers. Babies are illogical. Nobody is despised who can manage a crocodile. Illogical people are despised. Hence, babies cannot manage crocodiles.	7	CO1	K3
b)	Determine the PDNF and PCNF of $p \rightarrow [(p \rightarrow q) \wedge \sim(\sim q \vee \sim p)]$	7	CO1	K3
UNIT-II				
3.a)	Determine the number of ways of arranging 6 boys and 6 girls in a row. Also determine the number of arrangements in which i) All girls will be together. ii) No two girls will be together. iii) Boys and girls come alternatively.	7	CO2	K2
b)	i) Determine the term independent of x in the expansion of $(x^2 + \frac{1}{x})^{12}$ ii) Determine the coefficient of $x^5 y^{10} z^5 w^5$ in the expansion $(x + 7y + 3z + w)^{25}$	7	CO2	K2
(OR)				
4.a)	A cricket team of 11 is to be selected out of 14 players of whom 5 are bowlers. Determine the number of ways in which this can be done so as to include at least 3 bowlers.	7	CO2	K2
b)	Determine the number of integers between 1 and 250 which are divisible by any of the integers 2, 3, 5 or 7.	7	CO2	K3
UNIT-III				
5.a)	Let R denote a relation on the set of ordered pairs of positive integers such that $(x, y)R(u, v)$ if and only if $xv = yu$. Then establish that 'R' is an equivalence relation.	7	CO3	K2

b)	Define Poset and draw Hasse diagrams for the Posets (i) $[D_{12};/]$ (ii) $[\{2,3,5,30,60,120,180,360\};/]$ and (iii) $[P(A),\subseteq]$, where $A = \{1,2,3\}$	7	CO3	K2
(OR)				
6.a)	Establish that a Lattice 'L' is distributive iff $\forall x, y, z \in L \quad (x * y) \oplus (y * z) \oplus (z * x) \equiv (x \oplus y) * (y \oplus z) * (z \oplus x)$	7	CO4	K3
b)	Consider the Boolean polynomial $p(x, y, z) = x * (y \oplus z')$. If $B = \{0,1\}$, compute the truth table of the function $f: B_3 \rightarrow B$ defined by p . Also draw logical diagram.	7	CO4	K3
UNIT-IV				
7.a)	Determine the number of integral solutions for the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6$ and $x_5 \geq 0$.	7	CO5	K3
b)	Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 4^n$ for $n \geq 2$	7	CO5	K3
(OR)				
8.a)	i) Calculate $B(X) = \sum_{r=0}^{\infty} b_r X^r = \frac{1}{X^2 - 6X + 8}$ ii) Find the coefficient of X^{20} in $(X^3 + X^4 + X^5 + \dots)^5$	7	CO5	K2
b)	Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 0, n \geq 2$ by using generating functions.	7	CO5	K3
UNIT-V				
9.a)	Determine the Isomorphism between the following graphs. 	7	CO6	K3
b)	State and Prove Euler's formula for planar graphs.	7	CO6	K2
(OR)				
10.a)	Establish that a tree with 'n' elements has exactly 'n-1' edges.	7	CO6	K2
b)	Explain Kruskal's algorithm and using it find the minimal spanning tree for the following weighted graph 	7	CO6	K3

Course Code: B20HS2201				
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			R 20	
II B. Tech II Semester –MODEL PAPER				
MANAGERIAL ECONOMICS AND FINANCIAL ACCOUNTANCY (R 20)				
Common to CSE & IT				
Time : 3Hrs.			Max. Marks: 70	
Answer any one Question from EACH UNIT.				
All questions carry equal marks.				
S.No.	Question / Problem	CO	KL	M
	UNIT-I			
1.	Define Managerial Economics and Explain its nature and scope	CO1	K2	14
	OR			
2.	What do you mean by Elasticity of demand? Explain in detail about degrees of Price elasticity of Demand?	CO1	K2	14
	UNIT-II			
3.	Define Cost & classify the Elements of Cost?	CO2	K2	14
	OR			
4.	How do you calculate BEP? What are its Assumptions and Applications?	CO2	K3	14
	UNIT-III			
5.	What are Market Structures and explain the features of Perfect Competition?	CO3	K2	14
	OR			
6.	Why is pricing significant in the context of business? Describe any four pricing practices?	CO3	K2	14
	UNIT-IV			
7.	Describe about the Importance of Accounting and types of accounts	CO4	K2	14
	OR			
8.	From the following Trail Balance of Suresh as at December 31, 2013, prepare Trading, Profit and Loss Account for the year ended December 31, 2013 and a Balance Sheet as on that date:	CO4	K3	14

	Dr. (Rs.)	Cr. (Rs.)			
Purchases of materials	32,000				
Productive wages	13,000				
Sales		60,000			
Salaries	4,000				
Travelling expenses	1,000				
Carriage inwards	550				
Insurance	300				
Commission	650				
Rent and rates	1,000				
Cash in hand	350				
Cash at bank	5,550				
Repairs	600				
Sundry expenses	110				
Mortgage		6,100			
Buildings	8,000				
Machinery	3,000				
Furniture	1,000				
Stock on hand (1.1.2013)	11,500				
Capital		21,310			
Sundry debtors	9,000				
Sundry creditors		4,200			
	91,610	91,610			

	Adjust the following: (a) Prepaid rent Rs. 100 (b) Depreciate the following: • Buildings @ 10per cent per annum • Machinery @ 20 per cent per annum • Furniture @ 15 per cent per annum (c) Provide for bad debts Rs. 100 (d) Outstanding insurance Rs. 50 (e) Closing stock Rs. 12,000			
	UNIT-V			
9.	Explain about capital and the sources available for raising finance	CO5	K2	14
	OR			
10.	Explain about the concept and causes of depreciation. Evaluate the straight-line method and diminishing balance methods.	CO5	K2	14

Course Code: B20BS2202																
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)				R 20												
II B. Tech II Semester - MODEL QUESTION PAPER																
STATISTICS WITH R																
(for IT)																
Time: 3 Hrs.		Max. Marks:70														
Answer any one Question from Each Unit																
All questions carry equal Marks																
Q.No.	Questions	CO	KL	M												
1.a)	If X is a continuous random variable with probability density function given by $f(x) = \begin{cases} kx, & (0 \leq x < 2) \\ 2k, & (2 \leq x < 4) \\ -kx + 6k, & (4 \leq x < 6) \end{cases}$ find k and mean value of x.	CO1	K2	7M												
b)	Define moment generating function and write its properties.	CO1	K2	7M												
(OR)																
2.a)	Explain Data Frames in R to store Student sample data?	CO1	K2	7M												
b)	Explain arithmetic operations in R?	CO1	K2	7M												
3.a)	Fit a Poisson distribution to the set of observations <table><tr><td>x:</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>f:</td><td>122</td><td>60</td><td>15</td><td>2</td><td>1</td></tr></table>	x:	0	1	2	3	4	f:	122	60	15	2	1	CO2	K2	7M
x:	0	1	2	3	4											
f:	122	60	15	2	1											
b)	If X is a normal variate with mean 30 and standard deviation 5, determine the probabilities that i) $26 \leq x \leq 40$ ii) $x \geq 45$	CO2	K3	7M												
(OR)																
4.a)	Ten coins are thrown simultaneously. Find the probability of getting at least (i) Seven heads (ii) six heads. Write R function for the above problem.	CO2	K2	7M												
b)	Describe various probability distribution functions in R?	CO2	K3	7M												
5.a)	Explain different methods of sampling.	CO3	K2	7M												
b)	A population consists of five members 2,3,6,8 and 11. Consider all possible samples of size two which can be drawn without replacement from this population. Find (i) the mean of sampling distributions of means (ii) the standard deviation of the sampling distribution of means	CO3	K2	7M												
(OR)																
6.	Write about R-Probability distribution functions.	CO4	K3	14M												
7.a)	In a city A 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B 18.5% of a random sample of 1600 school boys had the Same defect. Determine whether the difference between the proportions significant?	CO5	K3	7M												

b)	In one sample of 8 observations from a normal population, the sum of squares of the deviations of the sample values from sample mean is 84.4 and in another sample of 10 observations it was 102.6. Test at 5% level whether the populations have the same variance.	CO5	K3	7M																						
(OR)																										
8.	Write R functions for applying t-test, F-test and chi-square tests.	CO5	K2	14M																						
9.a)	Calculate the rank correlation coefficient to the following data <table><tr><td>x</td><td>48</td><td>33</td><td>40</td><td>9</td><td>16</td><td>16</td><td>65</td><td>24</td><td>16</td><td>57</td></tr><tr><td>y</td><td>13</td><td>13</td><td>24</td><td>6</td><td>15</td><td>4</td><td>20</td><td>9</td><td>6</td><td>19</td></tr></table>	x	48	33	40	9	16	16	65	24	16	57	y	13	13	24	6	15	4	20	9	6	19	CO6	K3	7M
x	48	33	40	9	16	16	65	24	16	57																
y	13	13	24	6	15	4	20	9	6	19																
b)	Fit an exponential curve of the form $y = ae^{bx}$ to the following data <table><tr><td>x</td><td>0</td><td>0.5</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td></tr><tr><td>y</td><td>0.10</td><td>0.45</td><td>2.15</td><td>9.15</td><td>40.35</td><td>180.75</td></tr></table>	x	0	0.5	1.0	1.5	2.0	2.5	y	0.10	0.45	2.15	9.15	40.35	180.75	CO6	K3	7M								
x	0	0.5	1.0	1.5	2.0	2.5																				
y	0.10	0.45	2.15	9.15	40.35	180.75																				
(OR)																										
10.	Implement an R program for the linear regression model to the following data? <table><tr><td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td></tr><tr><td>y</td><td>15</td><td>19</td><td>23</td><td>26</td><td>30</td></tr></table>	x	5	10	15	20	25	y	15	19	23	26	30	CO6	K2	14M										
x	5	10	15	20	25																					
y	15	19	23	26	30																					

Course Code: B20IT2201					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech II Semester Regular Examinations					
COMPUTER ORGANIZATION					
(For IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
UNIT-I					
			CO	KL	M
1.	(a).	Explain about Fixed Point and Floating Point Representation with examples	CO1	K3	7M
	(b).	Explain about Booths multiplication algorithm with flow chart	CO1	K2	7M
(OR)					
2.	(a).	Explain about division algorithm with example	CO1	K3	7M
	(b).	Describe Bus Structures in basic digital computer system	CO1	K2	7M
UNIT-II					
3.	(a).	Describe Instruction cycle in computer system	CO2	K2	7M
	(b).	Design Arithmetic and Logic shift unit	CO2	K2	7M
(OR)					
4.	(a).	Explain about Bus and Memory Transfer	CO2	K2	7M
	(b).	Explain about Computer instructions	CO2	K2	7M
UNIT-III					
5.	(a).	Explain about General register organization with seven registers	CO3	K2	7M
	(b).	Expand the given statement in Three,Two,One,Zero Addresses A=(B+C)*(D+E)	CO3	K3	7M
(OR)					
6.	(a).	Describe the Addressing Modes i) Direct ii) Relative iii) Register with example	CO3	K3	7M
	(b).	Hardwired control Vs Micro programmed control	CO3	K2	7M
UNIT-IV					
7.	(a).	Discuss about Memory Hierarchy	CO4	K2	7M
	(b).	Explain about Asynchronous Communication interface with neat diagram	CO4	K2	7M
(OR)					
8.	(a).	Explain about Asynchronous Data transfer	CO4	K2	7M
	(b).	Explain Memory Mapping Techniques of Cache Memory	CO4	K2	7M
UNIT-V					
9.	(a).	Describe the Characteristics of Multiprocessors	CO5	K2	7M
	(b).	Explain about Array Processor.	CO5	K2	7M
(OR)					
10.	(a).	Describe Interconnection Structures with neat diagram	CO5	K2	7M
	(b).	Describe about Pipeline and vector processing	CO5	K2	7M

Code: B20IT2202					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)					R20
II B. Tech II Semester Regular Examinations					
JAVA PROGRAMMING					
(For AIDS & IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT - I					
1.	a).	Illustrate the concept of JVM with a Diagram.	CO1	K3	7M
	b).	Illustrate the differences between C,C++ and Java with a neat diagram	CO1	K3	8M
OR					
2.	a).	Illustrate the structure of a java program	CO1	K3	8M
	b).	Explain java buzz words in detail.	CO1	K3	7M
UNIT - II					
3.	a).	Write a java program to perform constructor overloading with an example	CO2	K2	8M
	b).	Explain class Declaration syntax in java and modifiers	CO2	K2	7M
OR					
4.	a).	Write a java program to perform method overloading.	CO2	K2	8M
	b).	Explain the keyword final how it is used in java	CO2	K2	7M
UNIT - III					
5.	a).	Explain polymorphism and its types. Construct a java program which Illustrates the functionality of method overloading and method overriding.	CO3	K3	8M
	b).	Write how Multiple Inheritance is possible in java with an example program.	CO3	K3	7M
OR					
6.	a).	Demonstrate an array? Write a java program to read an array of n elements and print them.	CO3	K3	8M
	b).	Write a java program to find second largest number in an array.	CO3	K3	7M
UNIT – IV					
7.	a).	Interpret the concept of packages in java.	CO4	K2	8M
	b).	Construct a java program that shows the functionality of creating a public class in an already existing user defined package.	CO4	K2	7M
OR					
8.	a).	Write a java program to create custom exception.	CO4	K3	8M
	b).	Illustrate 2 different ways of creating a thread in java with code.	CO4	K3	7M
UNIT - V					
9.	a).	Explain any 4 layout mangers in java with example code.	CO5	K2	8M
	b).	Write a java swing code to create a frame and add any five different components onto the frame.	CO5	K3	7M
OR					
10.	a).	Write a java program to explain event handling.	CO5	K2	8M
	b).	Explain JDBC Architecture and procedure to establish JDBC Database Connections.	CO5	K2	7M

Course Code: B20IT2201					
SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE (A)				R20	
II B. Tech II Semester Regular Examinations					
PRINCIPLES OF SOFTWARE ENGINEERING					
(For IT)					
Time: 3 Hrs.			Max. Marks:70		
Answer any one Question from Each Unit					
All questions carry equal Marks					
			CO	KL	M
UNIT - I					
1.	a).	Illustrate spiral model with neat diagram.	CO1	K3	7
	b).	Illustrate unified process.	CO1	K3	7
OR					
2.	a).	Categorize process assessment and improvement.	CO1	K4	7
	b).	Identify the elements in software engineering practice.	CO1	K3	7
UNIT - II					
3.	a).	Illustrate Extreme Programming (XP).	CO1	K3	7
	b).	Interpret any 4 agile process models.	CO1	K3	7
OR					
4.	a).	Classify the characteristics of negotiating requirements and validating requirements.	CO1	K3	7
	b).	Design the complete usecase model for the following system Vehicle Insurance Renewal System	CO1	K4	7
UNIT - III					
5.	a).	Identify the relationships in class diagram for online shopping .	CO2	K3	7
	b).	Design interaction diagrams for Online Book Sales With Mobile SMS. (Any 2)	CO2	K4	7
OR					
6.	a).	Categorize requirements modeling for WebApps.	CO2	K4	7
	b).	Design state chart diagram with your own example.	CO2	K4	7
UNIT - IV					
7.	a).	Categorize the concepts of design.	CO3	K4	7
	b).	Design component diagram for Railway Reservation System .	CO3	K4	7
OR					
8.	a).	Illustrate Architectural Styles.	CO3	K3	7
	b).	Interpret designing class based components.	CO3	K3	7
UNIT - V					
9.	a).	Categorize Testing Strategies for object oriented software.	CO4	K4	7
	b).	Interpret User Interface Analysis and Design.	CO4	K3	7
OR					
10.	a).	Interpret path testing with example.	CO4	K3	7
	b).	Categorize Testing Strategies for WebApps.	CO4	K4	7